CCCA Cell-State Analysis — Wu et al. 2020 (Acute Myeloid Leukemia)

GO / NO-GO GATES

Gate 1 — sample mixing & manifold continuity — GO

36 AML samples intermix across one continuous manifold; iLISI 9.4 of 23.6 ceiling (θ=10 peak, from 1.1 pre-Harmony), single connected component, one low-cycling compartment.

Gate 2 — embryonic module geography — GO (non-branching)

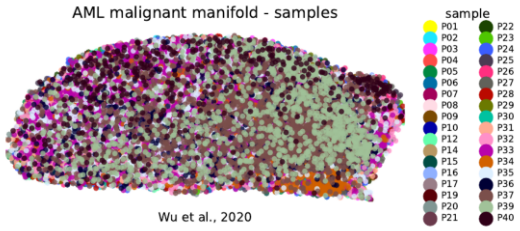
Single continuous Gast/pSom axis with organized cross-repression (smoothed r=-0.68) and a terminal Crest gradient (ρ=+0.43 vs pseudotime). Non-branching case per §1 — passes.

Gate 3 — hub Gast↔pSom cross-repression — NA

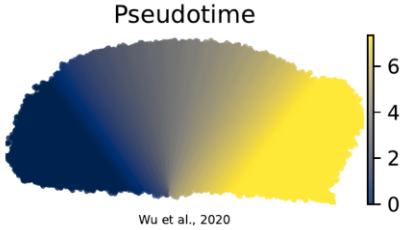
No bifurcation: the PPT tree's tips/fork carry near-identical signature means (geometric, not biological), so there is no co-expression hub or opposing arms to test. Branching-only gate — not applicable.

STEP FIGURES

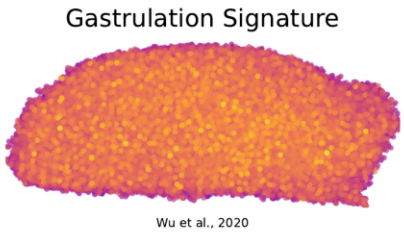
Gate 1 · sample integration



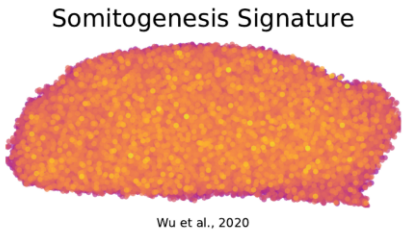
Trajectory · pseudotime (root→terminus)



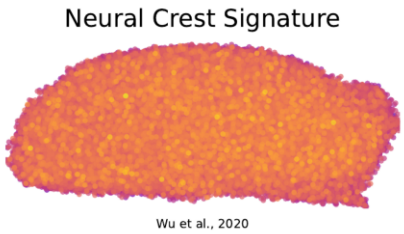
Gast (gastrulation)



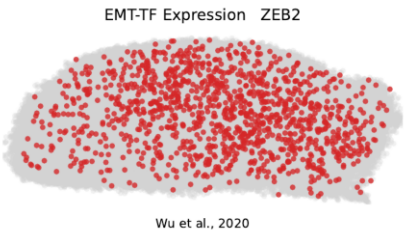
pSom (somitogenesis)



Crest (neural crest)



EMT-TF expression



Dataset. Acute myeloid leukemia · Microwell-seq · 96,881 malignant blasts from 36 bone-marrow samples.

Verdict. ✓ The embryonic-development pattern reproduces in its non-branching form — a single continuous gastrulation/somitogenesis axis under organized cross-repression, resolving into a terminal neural-crest gradient. No competing arms (Gate 3 NA). Geometry: Linear-pSom/Crest.

Parameter choices (all re-derived from this dataset).

Min-cell threshold:	202 (viable-range cliff 417→98, geo-mean); 36/38 samples kept
Integration:	Harmony on sample, θ=10 (iLISI sweep peak; drops at θ=20)
Embedding:	n_pcs=20, n_neighbors=30, spread=1.3
Geometry:	principal curve (ElPiGraph); fold-free ratio 2.90; continuity DC1 0.92
Root:	curve tip 0 — Gast/pSom-high, Crest-low (developmental origin)
Module means:	Gast +0.0019 · pSom +0.0066 · Crest -0.0042
EMT-TF readout:	ZEB2 (sole detectably-expressed factor; epithelial Snail/Zeb1/Twist silent in blasts)